Practice workbook · Explain how alveoli are adapted for gas exchange and effects of damage

Alveolar features for diffusion, differences between alveolar and outside air, and how emphysema reduces exchange.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ALVEOLI-GAS-EXCHANGE

1. Question 4 · part b.iii · 4 marks

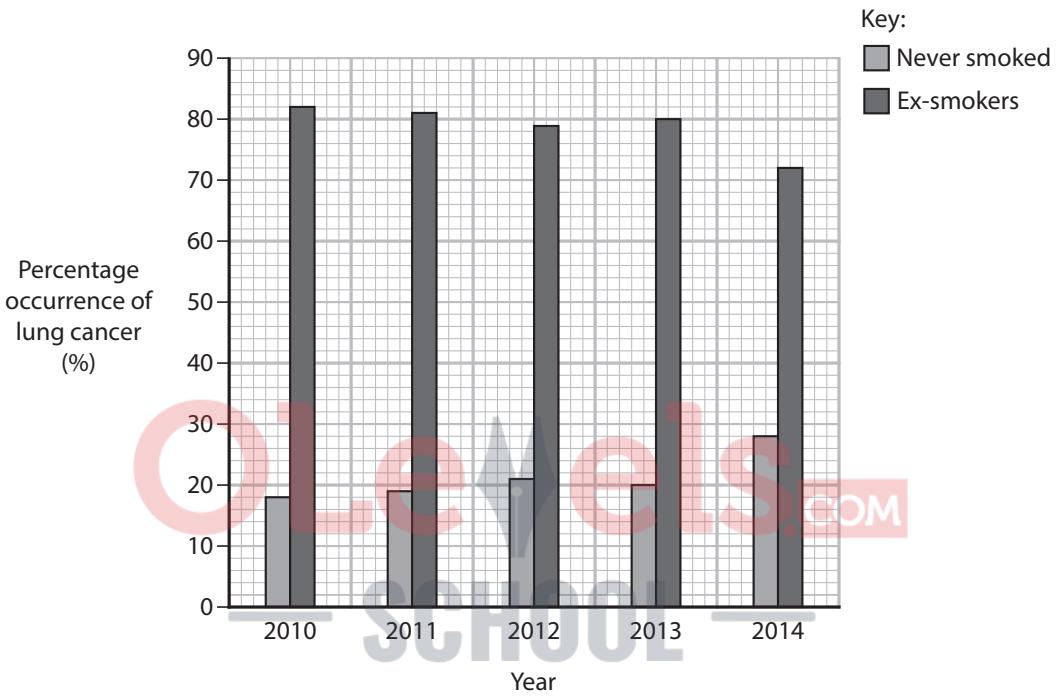
June 2025 Paper 2R

Worked-example retry · Attempt part b.iii only.

4 (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.



(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

(2)

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1. Question 4 · part b.iii · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.iii only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

(2)

ratio = ... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

(2)

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(iv) Name the substance in cigarette smoke that causes lung cancer.

(1)

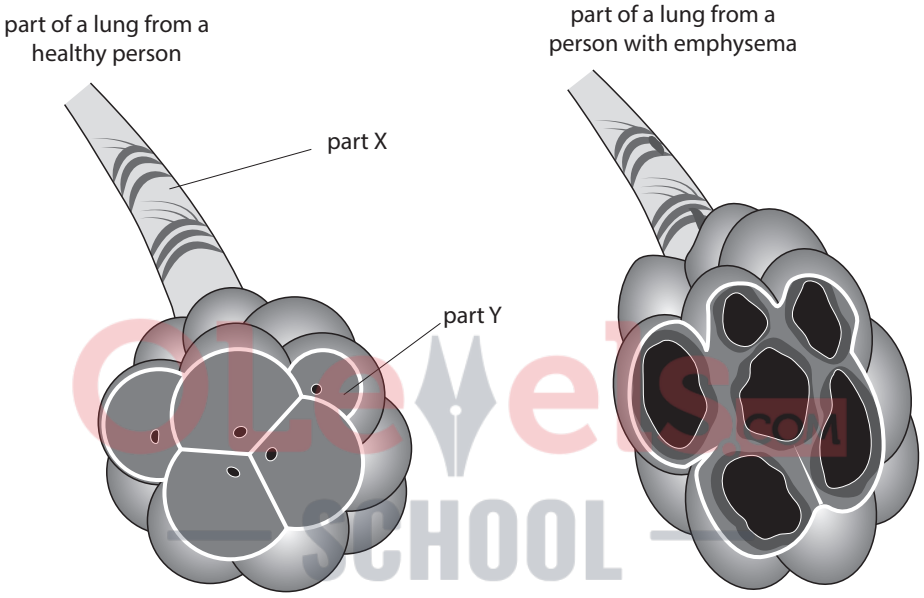
1. Question 4 · part b.iii · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.iii only.

(b) Emphysema is a lung condition that can be caused by cigarette smoking.

The diagram shows part of a lung from a healthy person and part of a lung from a person with emphysema.



(Source: © ttsz / Getty Images)

(i) Name part X.

(1)

(ii) Name part Y.

(1)

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1. Question 4 · part b.iii · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.iii only.

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(iii) Explain the effect that emphysema has on the health of a person.

(4)

(Total for Question 4 = 13 marks)

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Working and answer

2. Question 6 · part c.ii · 3 marks

June 2019 Paper 1

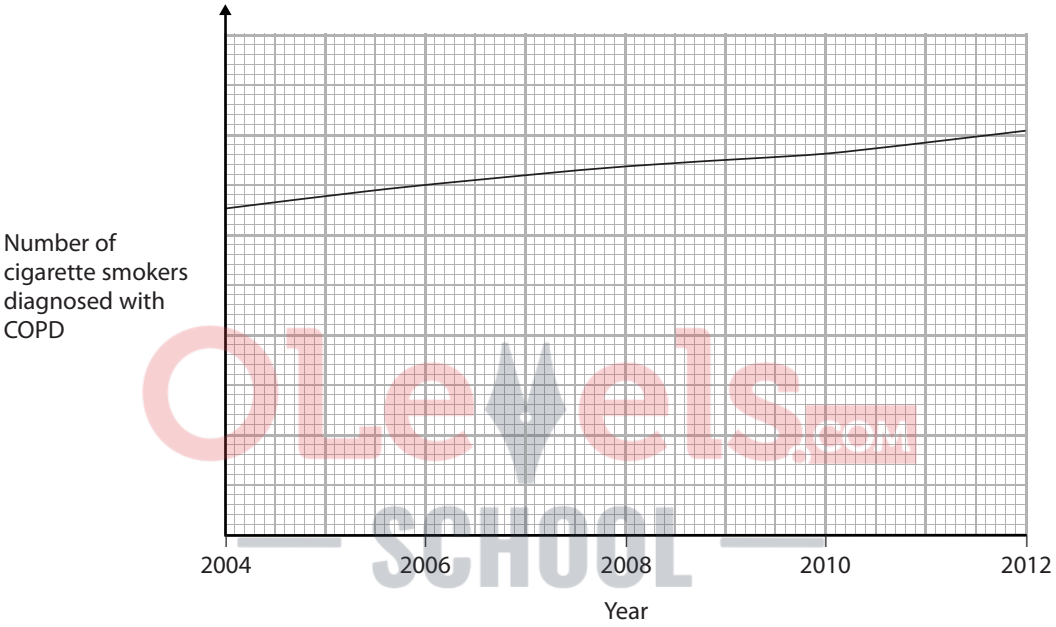
Further practice · Attempt part c.ii only.

6 (a) Chronic obstructive pulmonary disease (COPD) is the name given to a group of diseases that affect the breathing system.

These diseases include chronic bronchitis and emphysema.

Cigarette smoking is the main cause of COPD.

The graph shows the number of cigarette smokers diagnosed with COPD in the UK over a period of eight years.



(i) Describe the overall trend in cigarette smokers diagnosed with COPD.

(1)

(ii) Describe what further information is required to help form the conclusion that cigarette smoking is the only cause of COPD.

(2)

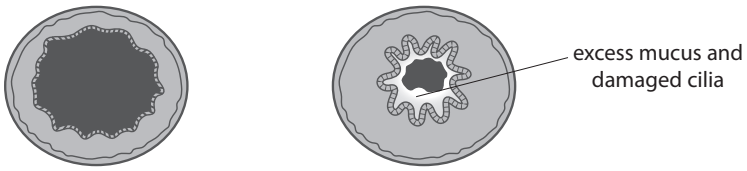
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2. Question 6 · part c.ii · 3 marks

June 2019 Paper 1

Further practice · Attempt part c.ii only.

(b) The diagram shows how chronic bronchitis affects the airways in the breathing system.



Explain how excess mucus and damaged cilia affect the breathing system of a person with chronic bronchitis.

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2. Question 6 · part c.ii · 3 marks

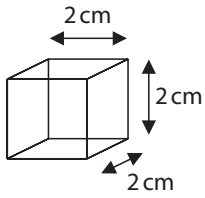
June 2019 Paper 1

Further practice · Attempt part c.ii only.

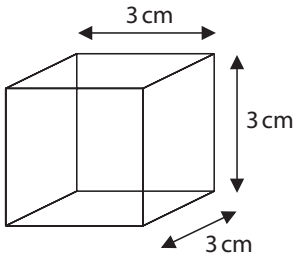
(c) The diagram shows models of two alveoli.

One model represents an alveolus from a person with healthy lungs.

The other model represents an alveolus from a person with emphysema.



Healthy



Emphysema

The table shows the surface area to volume ratio for a healthy alveolus.

	Surface area in cm^2	Volume in cm^3	Surface area to volume ratio
Healthy	24	8	3 : 1
Emphysema			

(i) Complete the table by giving the missing information.

(3)

(ii) Explain how the surface area to volume ratio of alveoli in the lungs of a person with emphysema will affect the normal function of body cells.

(3)

(Total for Question 6 = 11 marks)

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2. Question 6 · part c.ii · 3 marks · Working and answer

Working and answer

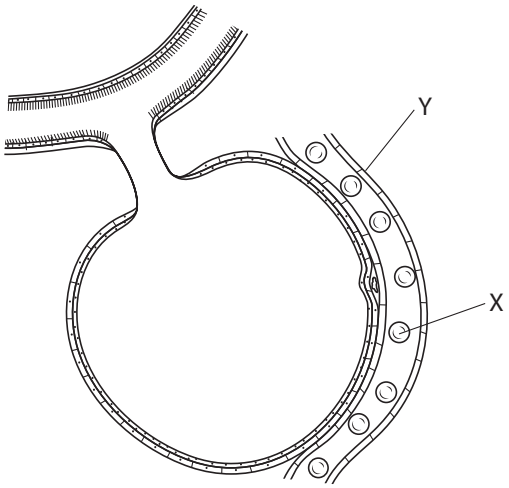
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3. Question 3 · part a.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

3 The diagram shows an alveolus and its blood supply.



(a) (i) What is the name of blood component X? **(1)**

- ☒ **A** plasma
- ☒ **B** platelet
- ☒ **C** red blood cell
- ☒ **D** white blood cell

(ii) State three ways in which air in the alveolus differs from air in the atmosphere. **(3)**

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3. Question 3 · part a.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

(iii) Give three features of alveoli that allow efficient gas exchange.

(3)

1 .....

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(b) Structure Y is a capillary.

Give two features in the diagram which show that structure Y is a capillary.

(2)

1 .....

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(c) Some people have a condition known as emphysema. One symptom of emphysema is the breakdown of elastic fibres in the lung tissue.

Suggest the effects that loss of elastic fibres have on a person's ability to breathe.

(2)

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(Total for Question 3 = 11 marks)

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3. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

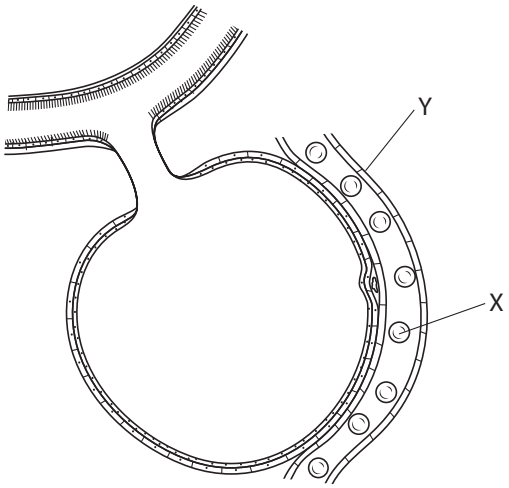
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4. Question 3 · part a.iii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.iii only.

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4. Question 3 · part a.iii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.iii only.

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(Total for Question 3 = 11 marks)

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4. Question 3 · part a.iii · 3 marks · Working and answer

Working and answer

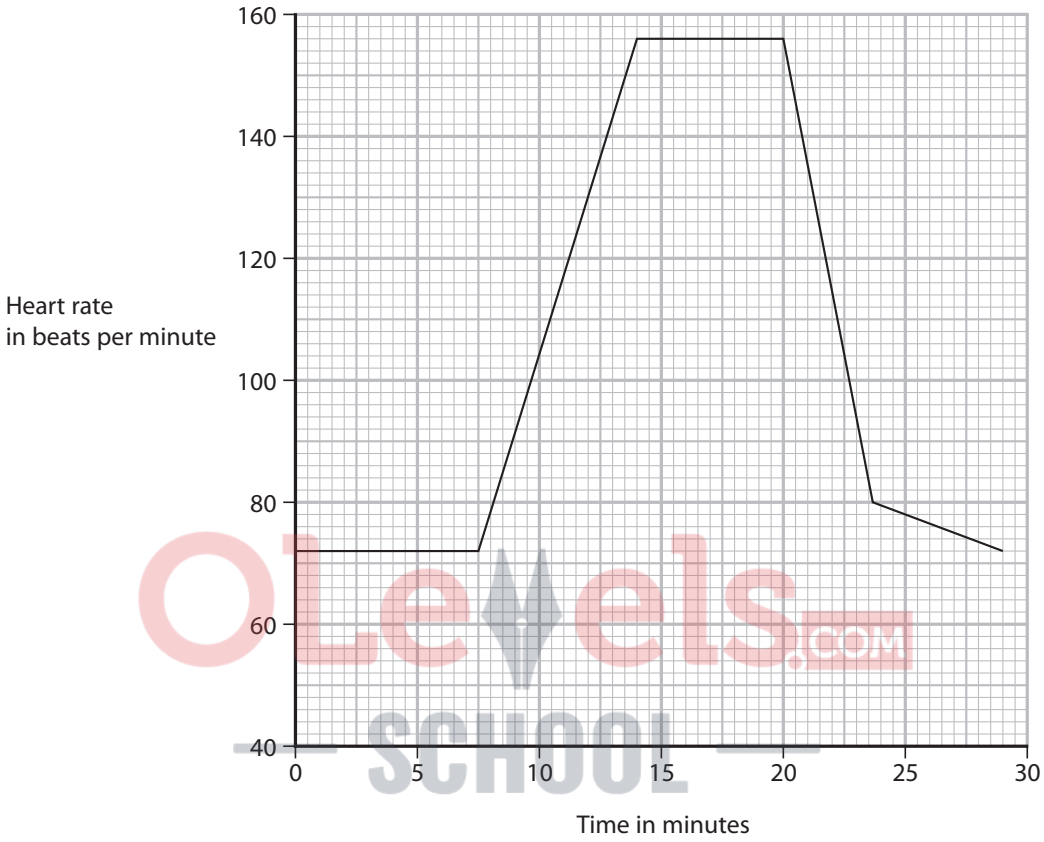
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5. Question 2 · part a.iii · 4 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

2 (a) The graph shows how a person's heart rate changes with exercise.



(i) Determine the difference between the maximum heart rate and the resting heart rate.

(2)

difference = ... beats per minute

(ii) Which of these is the time taken for this person to recover from the exercise?

(1)

- ☒ **A** 4 minutes
- ☒ **B** 6 minutes
- ☒ **C** 9 minutes
- ☒ **D** 15 minutes

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5. Question 2 · part a.iii · 4 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

(iii) Complete the passage about what happens in the body during exercise using the correct words.

(4)

During exercise ..... gas moves from the lungs into the blood by a process called ..... . This gas is needed by cells for ..... respiration which releases energy. Also during exercise, the amount of urine produced ..... as water is lost in sweat.

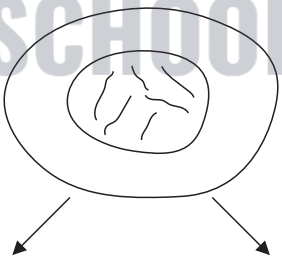
(b) (i) During exercise, body cells get damaged.

Body cells are replaced by a process called mitosis.

The diagram represents one body cell but shows only 6 of the 46 chromosomes.

Complete the diagram to show the new body cells produced from this cell by mitosis.

(2)



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5. Question 2 · part a.iii · 4 marks · Working and answer

Working and answer

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6. Question 2 · part a.ii · 2 marks

June 2023 Paper 1

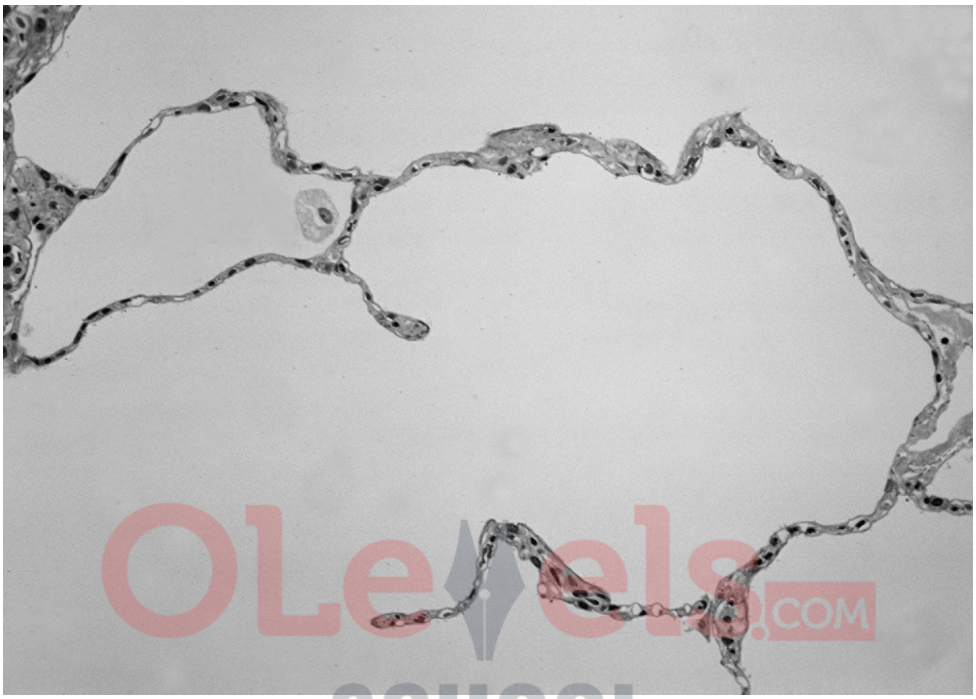
Further practice · Attempt part a.ii only.

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2 The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

(2)

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.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

(2)

1 .....

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2 .....

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6. Question 2 · part a.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.ii only.

(b) (i) Which is the percentage of carbon dioxide in the atmosphere?

(1)

- ☒ **A** 0.01
- ☒ **B** 0.04
- ☒ **C** 0.10
- ☒ **D** 0.50

(ii) The table shows the results obtained when a person at rest breathes in air that contains different percentages of carbon dioxide.

Percentage (%) carbon dioxide in air	0.02	1.60	3.00	6.00
Tidal volume in cm^3	500	700	1200	2100
Rate of breathing in breaths per minute	14	15	16	28

Calculate the volume of air in dm^3 breathed in per minute when the person breathes air containing 3.00% carbon dioxide.

[1 dm^3 = 1000 cm^3]

(2)

volume of air = ... dm^3

(iii) The air a person breathes in contains 20% oxygen. The body uses 20% of this oxygen.

Calculate the volume of oxygen used per minute when the person breathes air containing 1.60% carbon dioxide.

(3)

volume of oxygen = ... cm^3

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6. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

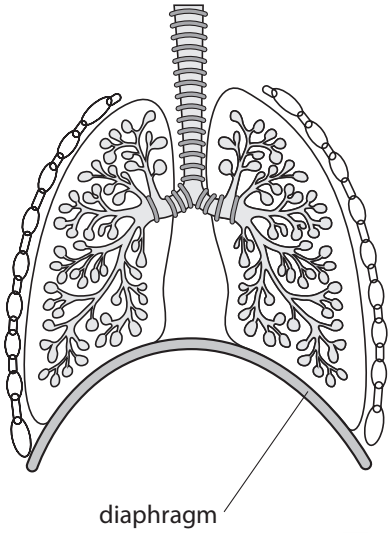
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7. Question 4 · part a.ii · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.ii only.

4 (a) The diagram shows the human lungs.



(i) Add the following labels on the diagram

(3)

- trachea
- bronchus
- alveoli

(ii) Give four features of the lungs that adapt them for gas exchange.

(4)

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7. Question 4 · part a.ii · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.ii only.

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(b) A spirometer can be used to test lung function.

(i) Describe how a spirometer is used to obtain values for tidal volume.

(4)

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(ii) Asthma is a condition that causes the tubes in the lungs to narrow.

A person without asthma has a vital capacity of 4600 cm^3.

People with severe asthma have a 30% lower vital capacity compared with a healthy person.

Calculate the vital capacity of a person with severe asthma.

(2)

vital capacity of a person with severe asthma = ... cm^3

(Total for Question 4 = 13 marks)

7. Question 4 · part a.ii · 4 marks · Working and answer

Working and answer

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